جامعة مصر للمعلوماتية
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Computer and Information Systems
Data Analysis Course

Does Using AI Tools (Like ChatGPT) Improve Learning or Reduce Effort?

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Introduction

Artificial intelligence has entered our daily lives whether we are ready or not. With usage among students and professionals growing rapidly around the world, the impact of AI tools on learning, productivity, and real understanding has become one of the most important questions of our time. This report uses real survey data to go beyond opinion and look at what people of different ages and backgrounds say about how AI affects their learning habits and effort.

To explore this, a structured survey was designed and distributed to a diverse group of respondents from school students to working professionals collecting honest, first-hand experiences with AI tools such as ChatGPT and Claude. Rather than sitting on the sidelines of this debate, this report puts real voices at the center : do people feel sharper and more capable because of AI, or are they quietly becoming dependent on a tool that thinks for them? The data will decide.

Research Question

Does the use of AI tools such as ChatGPT improve genuine learning, or does it simply reduce the effort individuals put into thinking independently?

Hypothesis

It is hypothesized that while AI tools make tasks faster and easier, they reduce the effort students and individuals put into thinking critically, leading to surface-level understanding rather than real learning

Population of Interest

The population of interest for this study includes school students, university students, and working professionals across different age groups, all of whom interact with AI tools in their academic or professional daily lives.

Sampling Method

Two sampling methods were combined in this study. First, convenience sampling: targeting people the researcher could easily reach through WhatsApp and social media. Second, snowball sampling: where those respondents passed the survey along to others, expanding the reach beyond the researcher's immediate circle. This combination was the most practical approach available, and it resulted in a diverse pool of respondents ranging from school students to working professionals.

Bias Identification:

Several potential sources of bias were identified in the design of this survey:

- Self-selection bias -> since participation was voluntary, people with strong opinions about AI were more likely to respond than neutral users.
- Social desirability bias -> some respondents may have answered in a way that sounds more acceptable rather than reflecting their true habits.
- Network bias -> since the survey was shared through personal networks, respondents may share similar backgrounds.
- Generational bias -> older respondents may be less familiar with AI tools, which could skew results toward lower usage simply due to less exposure.

To minimize these biases, the following steps were taken:

- The survey was kept fully anonymous to encourage honest answers.
- Diverse demographic groups were included across different ages and occupations.
- Rating scales were used rather than simple yes/no questions to capture more nuanced responses.
- Questions were phrased as neutrally as possible throughout the survey.

Survey Questions:

#	Survey Question
1	What is your current status? (School student / University student / Working)
2	Gender (Female / Male)
3	Age group (Under 18 / 18–22 / 23–30 / 30+)
4	How often do you use AI tools like ChatGPT, Claude…? (Never / Rarely / Sometimes / Often / Very often)
5	What do you mainly use AI tools for? (Studying/Coding / Writing/Summaries / General questions / I don't use AI tools)
6	On average, how much time do you spend using AI tools per day? (0 min / 10–30 min / 30–60 min / 1–3 hrs / 3 hrs+)
7	How much do AI tools help you understand topics? (1 = Not helpful, 5 = Extremely helpful)
8	After using AI tools, how often do you feel you actually learned something? (Never / Rarely / Sometimes / Often / Always)
9	Do you think AI tools make studying easier or just faster without real understanding?
10	How much do AI tools affect your study effort? (1 = I still work very hard, 5 = I rely heavily on AI)
11	After using AI tools, your productivity is: (Much worse / Worse / Same / Better / Much better)
12	Do AI tools make you finish tasks faster? (Yes / No / Sometimes)
13	If AI tools disappeared tomorrow, your performance would: (Improve / Stay the same / Get worse)
14	Do you think students rely too much on AI tools? (Strongly agree / Agree / Neutral / Disagree / Strongly disagree)

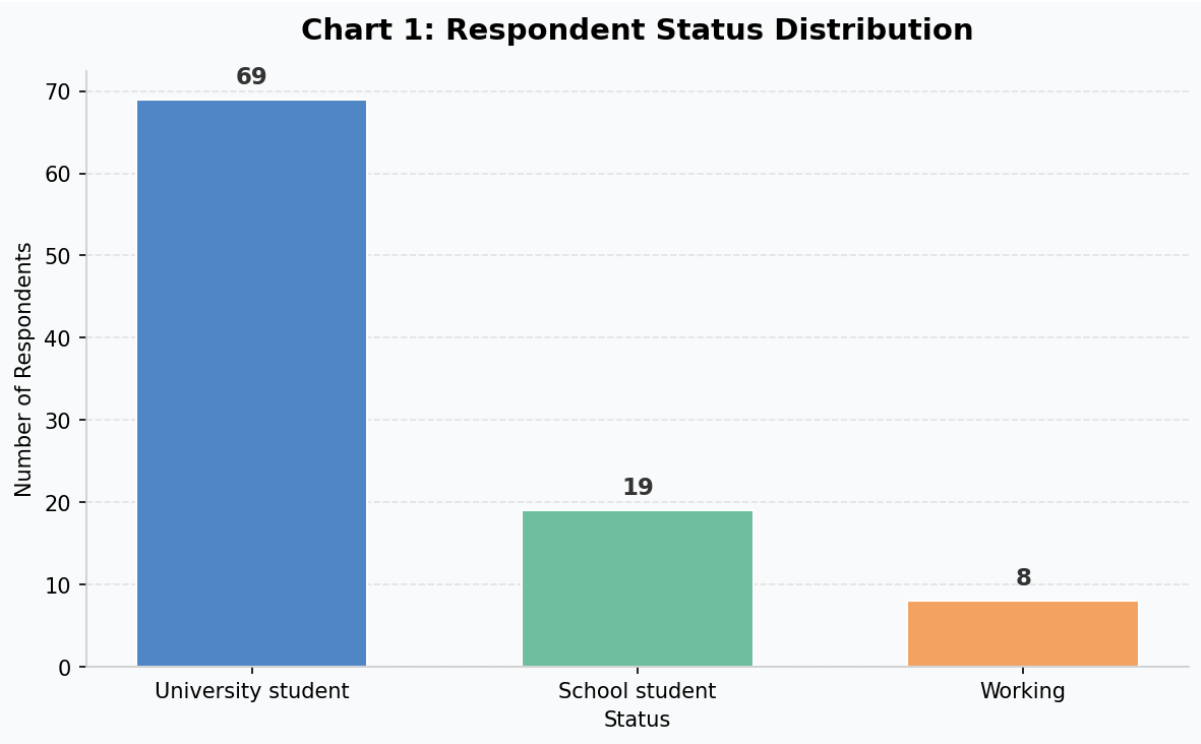
Online Survey Link: https://forms.gle/1FAIpQLSeVDuTad8mm0H1B6rBpV-HI2kcdvUZkQxOE9Bjb-dZSOAZRwg

Number of Samples Collected: 96 responses

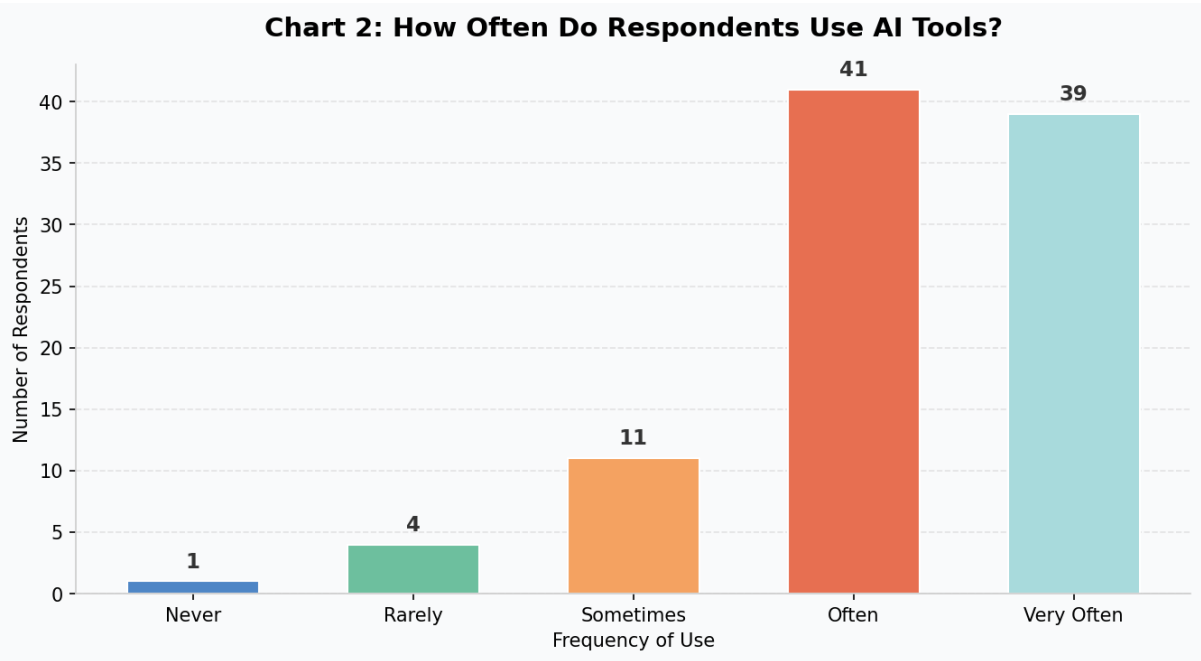
Analysis:

To analyze the data collected from this survey, we calculated basic descriptive statistics such as the mean, median, and mode for numerical rating questions. We also created visual representations of the data using charts and graphs to identify trends and patterns across the 96 responses collected.

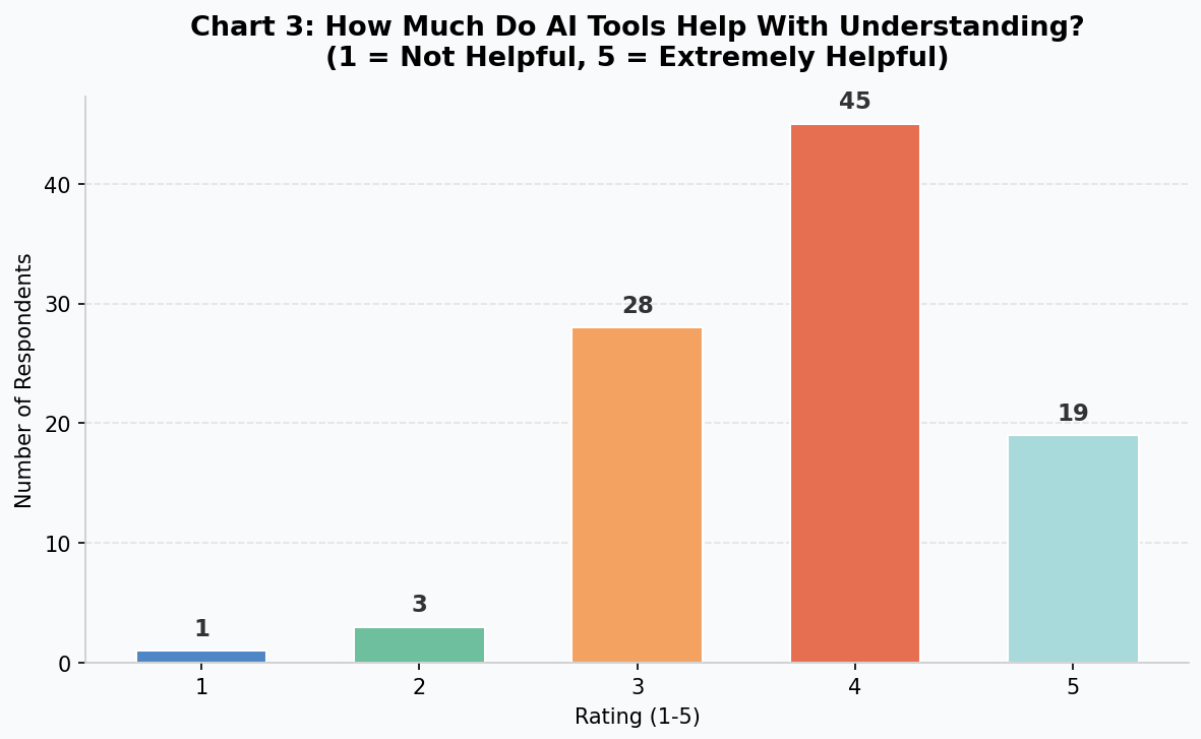
To quantify the survey results, descriptive statistics were calculated for the two numerical rating questions. For the question measuring how much AI tools help with understanding topics (rated 1–5), the **Mean was 3.81**, the **Median was 4.00**, and the **Mode was 4** indicating that the majority of respondents find AI tools quite helpful. For the question measuring how much AI tools affect study effort (rated 1–5, where 5 means heavy reliance), the **Mean was 2.68**, the **Median was 3.00**, and the **Mode was 3** suggesting that most respondents still put in moderate effort rather than fully relying on AI.



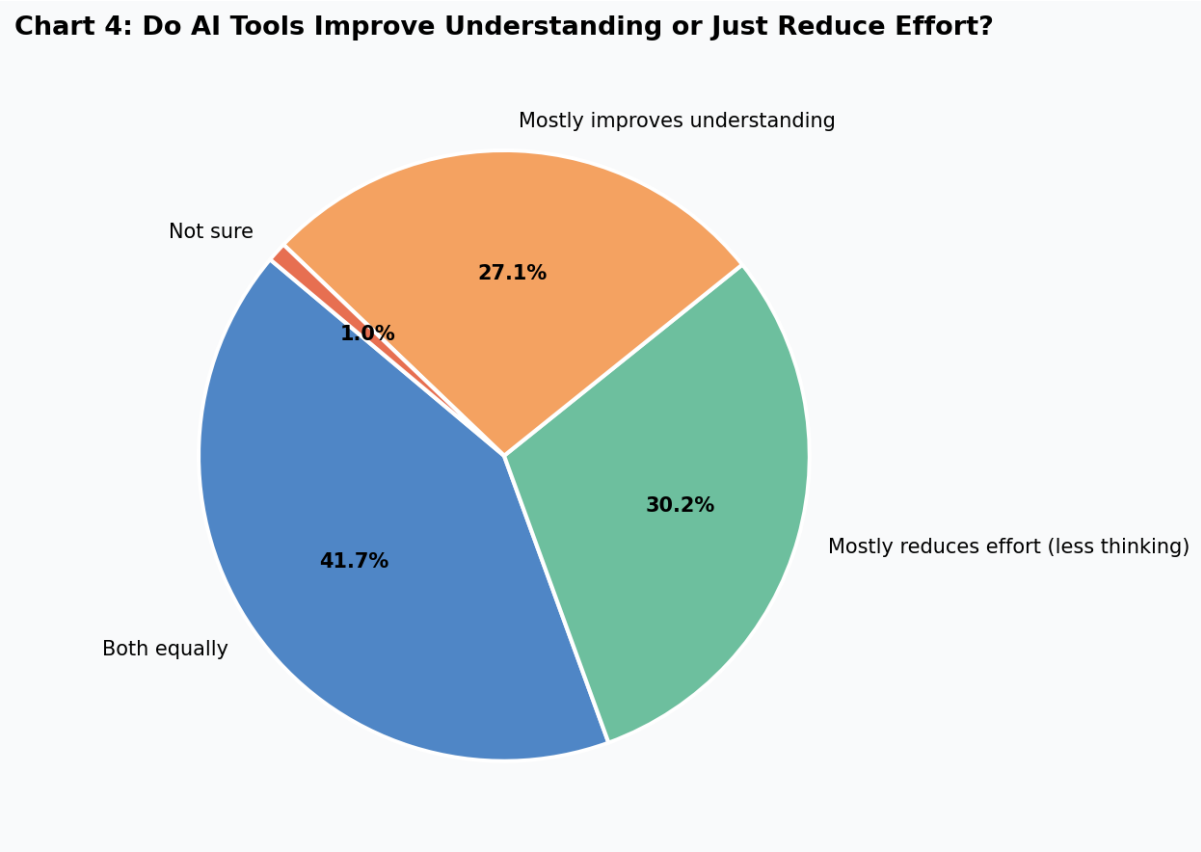
The majority of respondents (72%) are university students, followed by school students (20%) and working professionals (8%). This distribution reflects the natural reach of a student-led survey shared through academic networks, and means the findings are most representative of the university student experience with AI tools



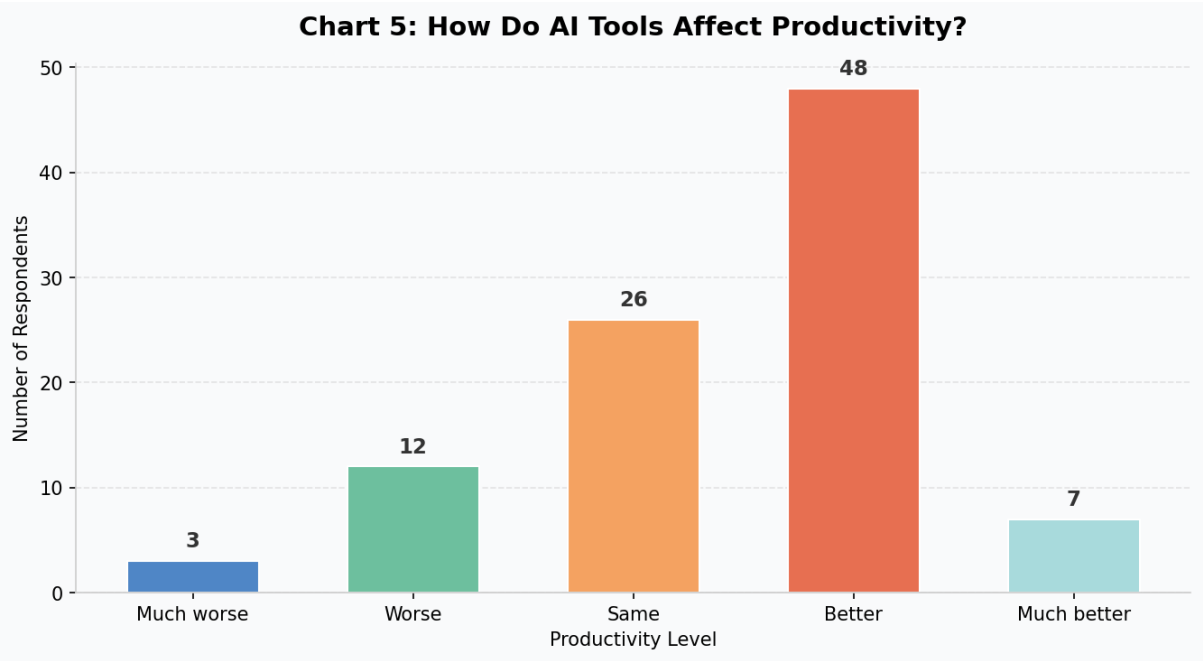
An overwhelming 83% of respondents use AI tools either "often" (3–5 times/week) or "very often" (daily), with only 1 respondent reporting never using them. This confirms that AI tools have become deeply embedded in the daily routines of students and professionals, making the question of their impact on learning highly relevant



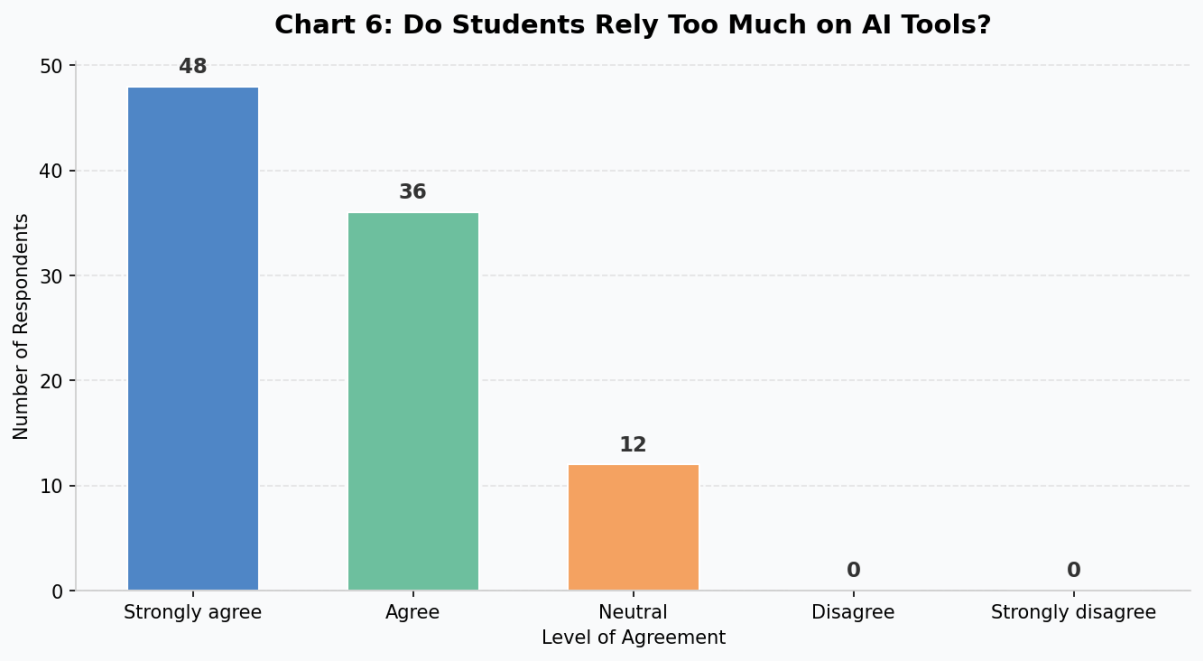
The most common helpfulness rating was 4 out of 5, given by 47%, with a mean rating of 3.85. This indicates that the majority of users find AI tools genuinely helpful for understanding topics, though the spread of responses suggests that the degree of benefit varies from one to another.



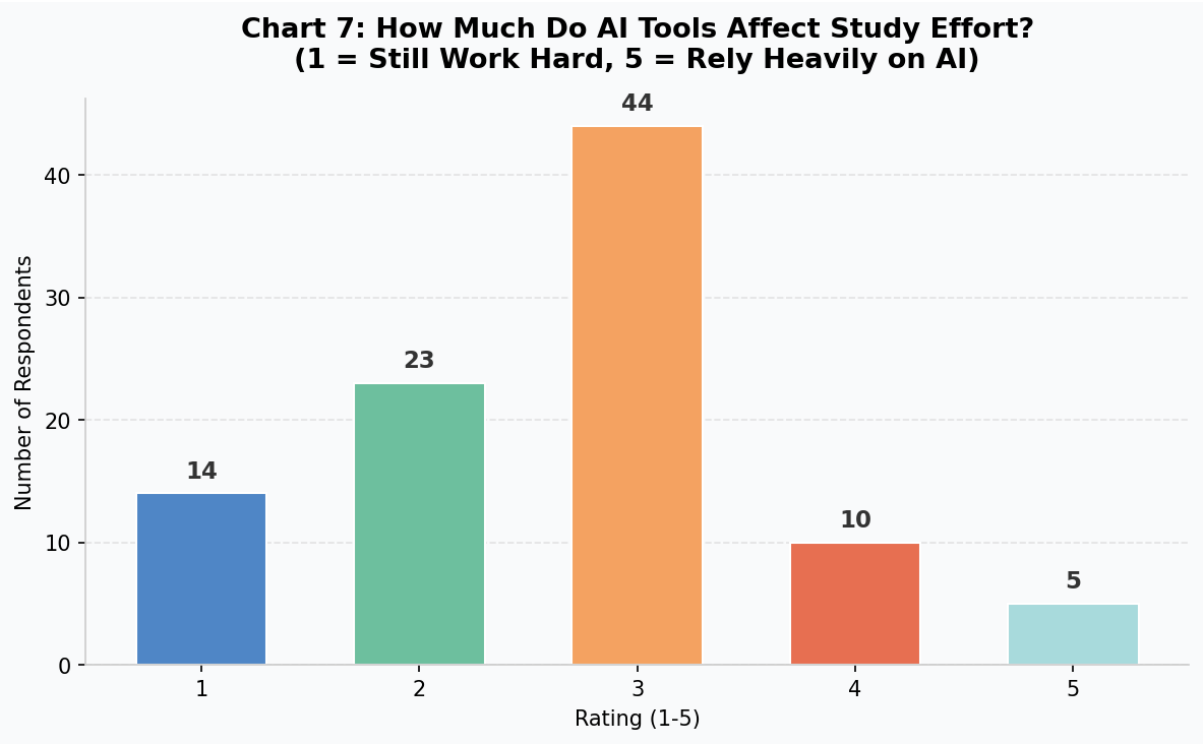
The largest group (41.7%) chose “Both equally,” indicating a mixed impact of AI tools. More respondents reported that AI mostly reduces effort (30.2%) rather than improves understanding (27.1%), supporting the idea that AI reduces critical thinking more than it enhances real learning.



50% reported their productivity as "Better", while only 15.6% reported it worse, This strongly suggests that AI tools have a positive effect on productivity and task completion, even if the nature of that productivity (deeper learning vs. faster output) remains debated.



A striking 88% of respondents either "Agree" or "Strongly Agree" that students rely too much on AI tools, with zero respondents selecting "Disagree" or "Strongly Disagree." This is one of the most telling findings of the study even heavy AI users themselves acknowledge the problem of over-dependence, suggesting awareness without necessarily a change in behavior.



The most common effort impact rating was 3 out of 5 (chosen by 46% of respondents), with a mean of 2.55. This indicates that most respondents still put in moderate effort when using AI tools rather than fully relying on them, though the distribution shows a clear lean toward reduced effort compared to working completely independently.

Conclusion

This study set out to investigate whether AI tools improve genuine learning or simply reduce the effort individuals put into thinking independently. Based on the data collected from 96 respondents, the findings partially support the hypothesis.

On one hand, the data shows clear productivity benefits that 50% of respondents reported better productivity after using AI tools, 75% confirmed that AI helps them finish tasks faster, and the mean helpfulness rating was 3.85 out of 5. These numbers suggest that AI tools do provide real value in day-to-day academic and professional tasks.

On the other hand, the data also reveals concerning patterns. When asked whether AI makes studying easier or just faster without real understanding, more respondents chose "Mostly reduces effort" than "Mostly improves understanding." Most strikingly, 88% of respondents agreed that students rely too much on AI tools which is a self-aware acknowledgment that even frequent users recognize the risk of over-dependence.

In conclusion, the data suggests that AI tools are genuinely helpful for productivity and speed, but they do come at a cost to independent thinking and deep learning for a significant portion of users. The hypothesis is therefore partially supported that AI tools do reduce critical thinking effort, but they also provide real learning benefits depending on how they are used. A more balanced and intentional approach to AI usage may be the key to maximizing its benefits while preserving independent thinking skills.

Any potential issues

Several limitations and hidden biases were discovered after analyzing the collected data. The most significant issue is the lack of demographic diversity because the majority of respondents were university students aged 18–22, which means the results cannot be fully generalized to working professionals or older age groups. Furthermore, the snowball sampling method, while effective for reach, created a concentration of respondents from similar academic backgrounds and social circles. A hidden bias that only became visible after data collection is that respondents who use AI heavily may have felt socially pressured to admit over-reliance, inflating the agreement rate on that particular question.